

SUJAL BOLIA

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EXPERIENCE

Amazon Web Services - S3 Replication

May 2025 – Present

Software Development Engineer

Vancouver, BC

- Designed and implemented the orchestration layer for a new replication workflow, including an optimized priority job store
- Implemented finalization logic that verified completion, coordinated concurrent updates, and enforced retry and failure policies
- Owned a redesign of state synchronization between two asynchronous distributed systems, eliminating persistent alert flapping, enabling closure of **60 outstanding incident tickets** and restoring reliable paging for genuine failures
- Delivered the first feature using an internal AI-agent orchestrator, uncovering issues and improving the framework
- Developed an **activity monitor** to detect stalled partitions, ensuring detection and mitigation before customer impact
- Deployed the service to a new AWS region, integrating dependent systems and validating cross-service reliability

IIT Bombay

July 2024 – Dec 2024

Machine Learning Engineer

Mumbai, MH

- Implemented a ResNet and BoTNet hybrid re-id model, achieving a state-of-the-art **mAP score of 85.6** with **97% fewer parameters**
- Designed and implemented a model-agnostic ML evaluation framework, **reducing evaluation time by 30%** and allowing researchers to assess models without modifying code
- Built a streamlined Docker image to containerize the ML framework, **reducing setup time by 40%**
- Fine-tuned a suite of computer vision XAI techniques to uncover hidden biases and enhance model transparency
- Integrated LLM-generated explanations with XAI outputs to improve the interpretability of computer-vision model results

EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Science in Computer Science

Sep 2020 – May 2024

PROJECTS

sc-extract | Rust, Image Processing, Concurrency, Compression Algorithms

[GitHub](#)

- Built a high-performance reusable library for decoding compressed binary asset formats and exporting image and data files
- Achieved over **10x speedup over existing implementations**, reducing a representative workload from **over 4 minutes to under 20 seconds** through efficient binary processing and parallelism
- Implemented endian-aware parsing, multiple pixel-format conversions, sprite-sheet reconstruction, geometric image transformations, and multi-format decompression
- Developed automated tests and cross-platform CI/release pipelines for Linux, macOS, and Windows

Brawlcord | Python, Rust, asyncio, SQLite, Redis

[GitHub](#)

- Built and operated an asynchronous text-based multiplayer game supporting **1,000+ players across 400+ communities**
- Designed a modular Rust framework for reusable gameplay mechanics, including concurrent battles and virtual-economy systems
- Used SQLite and Redis for persistent storage and low-latency caching

serenity-utils | Rust, Tokio, futures

[crates.io](#)

- Developed an async Rust crate to enhance the *serenity-rs* Discord API wrapper, with **over 10,000 downloads**
- Implemented customizable pagination menus, asynchronous input watchers with timeout for enhanced user interactivity, and custom model parsing for optimized data processing
- Provided over 10 fully generic helper models and functions to streamline development and improve code reusability

peerAdvice | TypeScript, React, Flask, Jinja, Firebase, Calendly

[GitHub](#)

- Collaborated with a team to build a peer-mentoring platform within 24 hours
- Implemented user authentication and Calendly scheduling, enabling students to discover mentors and request appointments

TECHNICAL SKILLS

Languages: Rust, Java, Python, Go, TypeScript, SQL, C/C++

Backend & Distributed Systems: Tokio, asyncio, PostgreSQL, Redis, Kafka, SQLx, FastAPI, Flask

Cloud & Infrastructure: AWS, Docker, GitHub Actions, CI/CD, Elasticsearch

AI & Machine Learning: PyTorch, scikit-learn, LLM Integration, AI Agents, Computer Vision, Explainable AI